the text-to-text format does not work well on regression tasks where the model is asked to generate the actual numerical value as the target instead of predicting the entire probability distribution. Raffel et al. (2020) resorted to only predicting the range to which a given value to be predicted belonged to instead of performing full regression withg T5.

How can we leverage T5 for highly accurate performance on predictive tasks, especially regression tasks? We propose LLM-Prop as an approach. LLM-Prop leverages T5 by directly adding a linear layer on top of its encoder for regression tasks. This linear layer can be composed with a sigmoid or softmax activation for classification tasks. Relying only on the T5 encoder reduces the total number of parameters by half which allows us to train on longer sequences and therefore incorporate more crystal information to improve predictive performance.

3.2.3 Input Processing

Removing stopwords. We removed stopwords from the text descriptions as this preprocessing step has been shown to improve performance on predictive tasks (Dieng et al., 2020; Niyongabo et al., 2020). We processed all publicly available English stopwords [3] and excluded them from the crystal text descriptions, except for digits and certain signs that may carry important information for crystal representation such as bond distances and angles.

Replacing bond distance with a [NUM] token and bond angle with an [ANG] token. Several works have shown that LLMs struggle to resolve contextual numerical information required for general common-sense reasoning (Wallace et al., 2019; Zhang et al., 2020; Geva et al., 2020; Thawani et al., 2021). For example, Zhang et al. (2020) replaced numbers in the training data with their scientific notation using the [EXP] token (i.e., *314.1* with 3141[EXP]2) to train a new BERT model which outperforms the original BERT on tasks that require numerical reasoning skills such as question answering. In our case, numbers not only add reasoning complexity but also increase the input sequence tokens since they are generally tokenized on a digit basis. We replace all bond distances and bond angles in the crystal text descriptions along with their units with [NUM] and [ANG] tokens, respectively. Those tokens are then added to the vocabulary as new tokens (i.e., we replace *3.03 Å* with [NUM] and *120 degrees or 120°* with [ANG]). While we are aware that this might also limit LLM-Prop to learn from the information related to bond distances and angles between atoms and molecules in the crystal structure, our results show that compressing the description as described—representing bond distances and angles with the same two special tokens described above—enables LLM-Prop to see more context in the text and achieve better performance.

Prepending a [CLS] token to the input. Devlin et al. (2018) showed that prepending a [CLS] token to every input, updating the embedding of that token together with the input tokens, and then using it for prediction improves predictive performance on downstream tasks. We used this same preprocessing step on T5, adding a [CLS] token in front of every input and also in the vocabulary, and then learning its

[3] https://github.com/igorbrigadir/stopwords